PROJECT DOCUMENTATION

Manufacturing Downtime Analysis

DEPI Graduation Project | Data Analysis Track

A concise reference covering project objectives, source data, data preparation, analytical modeling, DAX measures, and Power BI dashboard deliverables.

Project team	6 members
Dataset provider	Digital Egypt Pioneers Initiative (DEPI)
Workbook	Manufacturing_Line_Productivity.xlsx
Tools	Microsoft Power BI, Power Query, DAX
Deliverable	Six interactive Power BI dashboard pages

Consolidated project reference for the team repository.

1. Problem Statement & Project Objectives

Manufacturing operations can experience interruptions that reduce productive time and affect line performance. This project analyzes production and downtime records to make patterns in operating duration, downtime causes, and performance easier to explore.

Project objectives

• Examine production activity and compare actual operating time with standard or required time.
• Measure downtime duration and its contribution to actual production duration.
• Explore downtime causes, including factors classified as operator errors.
• Compare patterns across available dimensions such as date, product, flavor, size, shift, and operator.
• Present the analysis through interactive Power BI dashboards.

Project scope

The scope covers the supplied Excel workbook, Power Query preparation, a relational analytical model, DAX calculations, exploratory analysis in Power BI, and six dashboard pages. This document describes the implemented workflow and supplied project details; it does not claim operational improvements resulting from the analysis.

2. Business Requirements & KPI Framework

The report helps users review production performance, identify where downtime occurs, and compare causes and operating dimensions. KPI interpretations below follow the supplied DAX definitions.

Total Actual Duration	Sum of actual duration in minutes.
Total Standard Duration	Sum of standard duration in minutes.
Total Downtime Min	Sum of downtime minutes from Line downtime.
Downtime %	Total Downtime Min ÷ Total Actual Duration.
Min Required Time	Sum of related product minimum batch time for each production row.
Efficiency %	Min Required Time ÷ Total Run Time (min). This is the implemented project definition.
Operator Error Downtime	Downtime minutes filtered to factors marked Operator Error = Yes.
Actual Production (In Hours)	Total Production Time (Hours) – Total Downtime (In Hours).

Efficiency % is specifically the ratio of minimum required time to total run time in the supplied measure; it is not redefined here as another efficiency formula.

3. Data Understanding & Sources

The project uses Manufacturing_Line_Productivity.xlsx, supplied by DEPI. The workbook contains four source sheets.

Line productivity	Production batch records, including dates/times, operators, and duration-related fields.
Products	Product reference information, including product, flavor, size, and minimum batch time.
Downtime factors	Downtime descriptions and operator-error classification.
Line downtime	Batch-level downtime records and durations by factor; initially in a wide layout.

The production and downtime sheets provide operational records; Products and Downtime factors provide descriptive context. DimDate and DimBatches were added for shared date and batch analysis, and Z_Measures organizes measures.

4. Database Design & Data Modeling

The Power BI model follows a **Galaxy Schema (Fact Constellation Schema)**, with two fact tables connected to relevant dimensions.

Line productivity	Fact table containing production activity and duration data.
Line downtime	Fact table containing downtime records and durations.
Products	Dimension table describing products and minimum batch time.
Downtime factors	Dimension table describing downtime factors and operator-error classification.
DimDate	Date dimension created using DAX.
DimBatches	Distinct batch dimension created using DAX from both fact tables.
Z_Measures	Dedicated table used to organize DAX measures.

Relationships were created manually in Power BI. DimBatches combines distinct batch values from Line productivity and Line downtime. The model supports analysis across production and downtime records.

5. Data Flow & Analytical Workflow

The project follows a pipeline from the supplied workbook to the interactive Power BI report.

1. Source	DEPI Excel workbook containing four source sheets.
2. Preparation	Power Query: data types, column renaming, and unpivoting.
3. Modeling	Galaxy Schema with two fact tables and shared dimensions.
4. Calculation	DAX measures for production, downtime, and performance metrics.
5. Analysis & output	Exploratory analysis and six Power BI dashboard pages.

6. Data Cleaning & Transformation

The confirmed Power Query steps were limited to:
• Changing data types.
• Renaming columns.
• Unpivoting the Line downtime table to convert downtime-factor columns from a wide layout into a row-based structure.

No additional steps—such as duplicate removal, null-value treatment, or outlier handling—are claimed because they were not confirmed.

7. Exploratory Data Analysis (EDA)

Exploratory analysis was performed in Power BI. The report explores production and downtime through summaries and comparisons across available fields. Dashboard subjects include production overview, operator performance, downtime causes, line productivity, production trends, and product/batch performance.

The documentation does not attribute EDA to Python, Excel, or external statistical packages.

8. Data Analysis & Insights

The analysis is organized around operational questions that can be explored through the model's measures and report filters.

Production performance	Actual and standard duration, run time, minimum required time, and calculated efficiency.
Downtime performance	Total downtime, downtime events, average downtime, and downtime as a share of actual duration.
Downtime causes	Contribution of downtime factors and downtime associated with operator-error factors.
Operator performance	Differences in operator-related downtime, errors, and batch processing time.
Productivity patterns	Production and downtime comparisons by product, flavor, size, shift, and date fields.

The dashboards are analytical outputs, not evidence that an operational intervention has been implemented. Specific numeric findings are not stated here; this document focuses on analytical coverage and supplied measures.

9. Dashboard Design & Visualization

The Power BI report contains six dashboard pages. Descriptions below summarize the subjects shown in the supplied screenshots; they are descriptive labels and may differ from exact page titles.

1. Production Overview	Production and operator overview, including duration and efficiency measures.
2. Operator Performance	Operator downtime/errors and batch processing time.
3. Downtime Analysis	Downtime totals, event counts, average downtime, and causes.
4. Line Productivity & Downtime	Downtime comparisons by flavor, size, shift, and production trends.
5. Production Overview & Trends	Batch and product totals, production time, and daily production trend.
6. Product & Batch Performance	Average batch time, efficiency percentage, and batches by product/flavor.

The report supports interactive exploration through Power BI visuals and available filters/slicers. The six pages provide complementary views of production activity, downtime, operators, and product-level performance.

Appendix A. Selected DAX Measures

The following reference lists calculations supplied for the project. Measure and column names are preserved as provided.

Total Actual Duration

```
Total Actual Duration = SUM('Line productivity'[Actual Duration (Min)])
```

Total Standard Duration

```
Total Standard Duration = SUM('Line productivity'[Standard Duration (Min)])
```

Total Downtime Min

```
Total Downtime Min = SUM('Line downtime'[Downtime (In Mintues)])
```

Downtime %

```
Downtime % = DIVIDE([Total Downtime Min], [Total Actual Duration], 0)
```

Total Run Time (min)

```
Total Run Time (min) = SUM('Line productivity'[Duration_Min])
```

Min Required Time

```
Min Required Time = SUMX('Line productivity', RELATED(Products[Min batch time]))
```

Efficiency %

```
Efficiency % = DIVIDE([Min Required Time], [Total Run Time (min)])
```

Total Production Time (Hours)

```
Total Production Time (Hours) = SUM('Line productivity'[Total Production Time (Hours)])
```

Total Downtime (In Hours)

```
Total Downtime (In Hours) = SUM('Line downtime'[Downtime (In Hours)])
```

Actual Prodction (In Hours)

```
Actual Prodction (In Hours) = [Total Production Time (Hours)] - [Total Downtime (In Hours)]
```

Total Downtime (Minutes)

```
Total Downtime (Minutes) = SUM('Line downtime'[Downtime (In Mintues)])
```

Operator Error Downtime

```
Operator Error Downtime = CALCULATE([Total Downtime (Minutes)], 'Downtime factors'[Operator Error] = "Yes")
```

Total Downtime Min and Total Downtime (Minutes) both sum the same downtime-minute column and appear to be parallel/duplicate measures. Original names and source-column spelling are retained.

10. Testing & Validation

Detailed testing procedures, validation checks, and test outcomes were not supplied. Therefore, this document does not claim that specific validation tests passed. Before publishing or using the report for operational decisions, the team should verify relationship behavior, filter propagation, measure results against source records, and handling of blank or missing values.

Project Deliverables & Summary

The deliverable is a Power BI report containing six interactive dashboard pages, supported by a prepared data model and DAX measures. Workflow: DEPI Excel workbook → Power Query preparation → Galaxy Schema model → DAX calculations → exploratory analysis and six Power BI dashboards.

This consolidated document summarizes the ten requested documentation areas using confirmed project details and avoids unverified findings or implementation claims.